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Modern Deep Learning Framework Using Temporal Fusion Transformer to Optimize Sequestration Efficiency and Oil Production in CCUS Projects

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Abstract

Optimizing Water-Alternating-Gas (WAG) processes is critical for maximizing both oil recovery and carbon sequestration efficiency in CO₂ Enhanced Oil Recovery (EOR) projects. Key design parameters include injection rate, the gas-to-water volume ratio (WAG ratio), and throughput—defined as the pore volume of CO₂ injected per day. Conventional optimization using simulation models can be cumbersome due to the vast design space and high uncertainty. In this study, we apply a deep learning model—the Temporal Fusion Transformer (TFT)—for multivariate forecasting of oil production and CO₂ sequestration efficiency across a range of WAG scenarios using field data from six legacy CO₂ EOR projects.

Digitized monthly production and injection data were collected from six U.S. fields representing diverse depositional environments. For each field, historical waterflood trends were isolated using exponential decline, and the incremental CO₂ response was modeled across various WAG ratios. The TFT model was trained on this multivariate time-series data to forecast one year of production response for both oil and CO₂ production. Hyper parameter optimization was used to optimize model inputs. The model achieved high predictive accuracy with R² values exceeding 0.87 and robust MAE factors.

The forecasts enabled actionable operational recommendations. For instance, adjusting the WAG ratio from 0.9 to 1.7 in the Denver Unit increased CO₂ retention by 48% and utilization efficiency by 38%, while reducing the WAG ratio in East Vacuum from 0.7 to 0.2 yielded a retention improvement of 47%. These optimizations directly support enhanced carbon abatement by reducing CO₂ recycling and improving sequestration permanence. To our knowledge, this is the first application of the TFT model for real-field CO₂ EOR forecasting and WAG strategy optimization. The approach offers a scalable, data-driven alternative to simulation workflows, aligning EOR operations with climate and sustainability targets.

Introduction

Global climate commitments and net-zero ambitions have intensified the focus on technologies that can simultaneously enhance hydrocarbon recovery and sequester carbon dioxide (CO₂). Carbon Capture, Utilization, and Storage (CCUS) represents one of the most mature, technically feasible, and economically viable decarbonization pathways available to the oil and gas industry [1]. Within this context, Enhanced

Oil Recovery (EOR) using CO₂—particularly in Water-Alternating-Gas (WAG) injection schemes—has gained renewed interest for its potential to maximize incremental oil recovery while simultaneously storing significant quantities of CO₂ underground.

CO₂ EOR has been widely applied in the Permian Basin and other mature reservoirs for decades [1]. Yet, optimization of WAG operations remains an ongoing technical and economic challenge. The ratio of gas to water in WAG cycles—commonly represented as the gas-water ratio (GWR)—has a significant influence on reservoir sweep efficiency, mobility control, breakthrough timing, and ultimately on both oil production and CO₂ retention [2]. Traditional approaches for optimizing GWR rely on numerical reservoir simulations, empirical decline curve analysis, or rule-of-thumb adjustments informed by operational experience. While effective to a degree, these methods are computationally expensive, time-consuming, and often lack the flexibility for short-term scenario testing.

Recent advances in machine learning (ML) and deep learning (DL) offer new opportunities to transform CO₂ EOR optimization workflows. Time series forecasting models—particularly deep neural networks—can learn complex, nonlinear relationships from historical data without explicit assumptions about reservoir behavior. Among these, the Temporal Fusion Transformer (TFT) has emerged as a powerful architecture for multivariate forecasting tasks, combining the strengths of recurrent and attention-based networks [3]. TFT integrates static and time-varying inputs, handles missing data robustly, and offers interpretability via attention weights—making it suitable for data-rich but complex subsurface problems.

Despite the demonstrated success of deep learning in power forecasting, production scheduling, and process optimization, applications to CO₂ EOR forecasting remain limited. Most prior efforts in CO₂ EOR forecasting have focused on simpler ML models such as Random Forests, XGBoost, or LSTM networks [4] [5] [6] [7] [8] [9]. These models, while effective for certain tasks, lack the temporal richness and attention mechanisms of TFT. Moreover, few studies have attempted to link ML forecasts with operational metrics such as CO₂ utilization (MCF/BBL) and retention (%), both of which are critical for balancing economic returns and environmental stewardship.

In this study, we present a novel, data-driven framework that leverages TFT to forecast 12 months of CO₂ and incremental oil production across six legacy CO₂ EOR fields in the Permian Basin: East Vacuum (EV) [10] [11] [12], Denver Unit (DU) [1] [13] [14], Wasson San Andres (WSA) [13], Seminole San Andres Unit (SSAU) [15], SACROC [16] [17], and Rangely Weber Sand (RWS) [18] [19]. These fields were selected due to their long production histories, public data availability, and diverse operational characteristics. Production and injection data were digitized from historical publications and technical papers using Automeris.io [17] and preprocessed to isolate the incremental oil response to CO₂. For each field, the waterflood trend was extrapolated using an exponential decline curve to estimate baseline performance; the difference was attributed to CO₂ injection response—a method consistent with prior analytical approaches [21].

To evaluate short-term operational strategies, five different WAG scenarios were created per field by varying the GWR. These scenarios were inputted into the TFT model, which was trained on historical data and used to forecast monthly oil and CO₂ production. Performance metrics including RMSE, MAE, and R² were calculated to assess model accuracy. Additionally, we calculated CO₂ throughput, utilization, and retention efficiency using assumed formation volume factors (FVF) of 0.5 for CO₂ and 1.0 for water as these values are common in Permian basin CCUS fields [13] [14]. This allowed us to compute a novel efficiency score (retention/utilization) to rank WAG scenarios by both economic and environmental merit.

Our results demonstrate that moderate WAG ratios (~1.5–2.5) offer the best trade-off between incremental oil recovery and CO₂ retention. The approach presented here enables rapid scenario screening without full-scale simulation, making it highly applicable for annual planning and operational optimization. To the best of our knowledge, this is the first study to apply a Temporal Fusion Transformer across multiple legacy EOR fields for CCUS optimization. This work not only advances the application of AI in reservoir engineering but also contributes a replicable framework for accelerating digital CCUS strategies.

Field and Data Description

This study focuses on six legacy CO₂ EOR fields in the United States with mature operational histories and publicly available production and injection data: East Vacuum (EV), Denver Unit (DU), Wasson San Andres Field (WSA), Seminole San Andres Unit (SSAU), SACROC, and Rangely Weber Sand (RWS). These fields represent a diverse range of reservoir architectures, injection strategies, and WAG implementations, offering a robust testbed for developing and validating short-term machine learning-based forecasting models.

Field Selection and Characteristics

The selected fields span the Permian Basin and other major CO₂ EOR hubs, encompassing a range of heterogeneity, operational practices, and injection histories:

- Denver Unit (DU): The Denver Unit in the Wasson Field is a major San Andres carbonate reservoir under CO₂ EOR since 1983, following pilot testing in 1978. It features strong surveillance programs and variable WAG ratios. The field is currently operated by Occidental Petroleum [1] [13] [14].
- SACROC: The SACROC Unit in the Midland Basin began CO₂ injection in 1972, transitioning from immiscible to miscible floods over time. It targets the Pennsylvanian Canyon Reef with decades of EOR experience. Kinder Morgan is the current operator [2] [17].
- Wasson San Andres Field (WSA): The WSA represents a combination of units (e.g., Willard, Bennet Ranch, DU) in the San Andres formation. Discovered in 1936, it includes thick carbonate with extensive WAG-based EOR development since the 1980's [13].
- Seminole San Andres Unit (SSAU): SSAU includes complex carbonate facies and a well-developed Residual Oil Zone. It has a long CO₂ EOR history with consistently high WAG ratios and known field-wide geological heterogeneity [15].
- East Vacuum (EV): East Vacuum is a smaller, data-rich field with phased CO₂ injection. It serves as a valuable site for short-term CO₂ performance evaluation [10] [11] [12].
- Rangely Weber Sand (RWS): RWS lies in the Paradox Basin and is geologically distinct with a siliciclastic Weber Sand reservoir. Its behavior under CO₂ EOR differs from carbonate analogs, making it a valuable contrast field [18] [19].

Data Acquisition and Digitization

Historical production and injection data were digitized using Automeris.io [17] [20], an open-source graphical digitization tool, from publicly available technical sources and additional literature and field summaries from DOE reports and technical presentations were also referenced to contextualize the data and validate trends [1] [2] [10] [11] [12] [13] [14] [15] [16] [17] [18] [19].

Where necessary, data were smoothed using a moving average filter to minimize digitization noise. Missing values or inconsistencies were handled through interpolation or masking during model training. All time series were aligned to a common index (month since CO₂ start), allowing for a standardized forecasting horizon across fields. See examples of digitized DU data [1] [13] [14] in Figure 1, while EV field production data, shown in Figure 2, was a combination of digitized data and publicly sourced data [10] [11] [12].

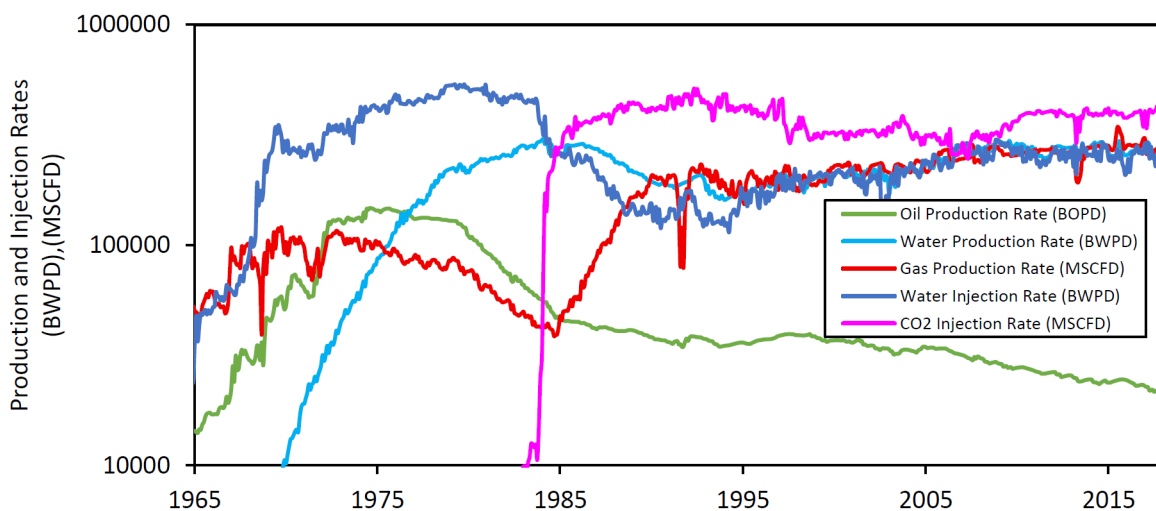


Figure 1—Denver Unit Production and Injection plot based on data digitized from public sources [1] [13] [14]

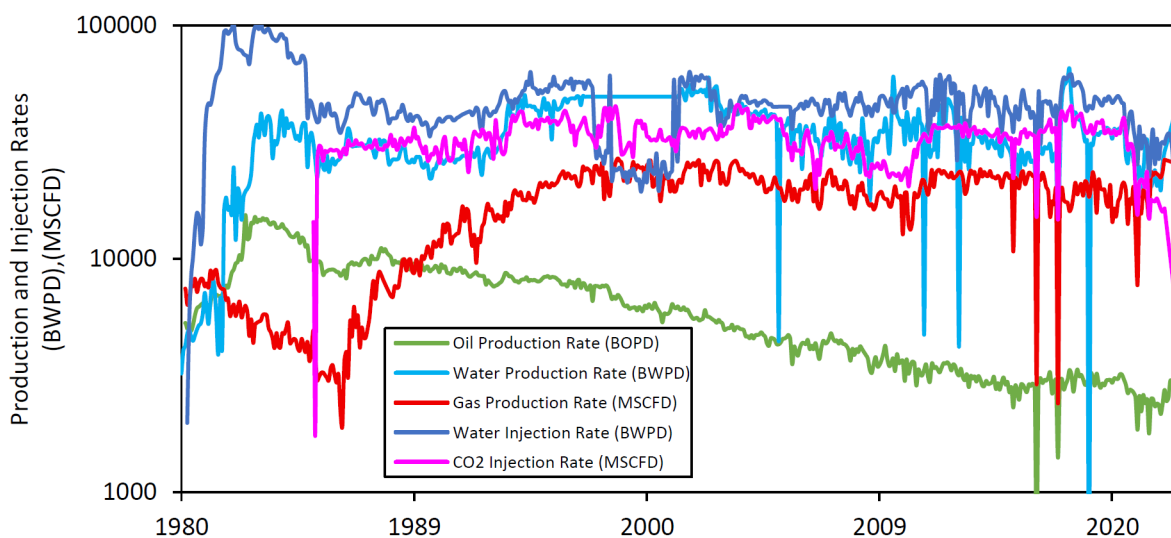


Figure 2—EV Production and Injection plot based on data digitized from public sources [10] [11] [12]

Parsing Incremental CO₂ Response and Production

To isolate the oil production response attributable to CO₂ injection, the waterflood performance prior to CO₂ start was extrapolated using exponential decline curve analysis, following a common analytical practice for baseline estimation [21]. The deviation from this trend during the CO₂ injection phase was attributed to incremental recovery from CO₂. This method is particularly important in mature fields where the transition from waterflood to CO₂ injection does not always exhibit a clean production breakpoint.

For SACROC, the start of the analysis window was aligned with the commencement of miscible CO₂ injection to exclude prior immiscible or waterflood effects. The parsed EOR response was used as the target variable in the TFT forecasting model, ensuring that predictions represent CO₂-specific incremental oil rather than total field production.

To isolate CO₂ production response, the gas-oil-ratio (GOR) prior to CO₂ injection was taken as a baseline and held constant throughout the CO₂ production period to extrapolate hydrocarbon gas production which enabled parsing out CO₂ production.

Derived Metrics

To assess the performance of different WAG scenarios, several key operational metrics were computed:

- CO₂ Utilization (MCF/BBL): Total CO₂ injected divided by incremental oil produced.
- CO₂ Retention (%): Defined as $100 \times (\text{CO}_2 \text{ injected} - \text{CO}_2 \text{ produced}) / \text{CO}_2 \text{ injected}$.
- Gas-Water Ratio (GWR): Defined as MCFPD / BWIPD, used to characterize WAG regime.
- Throughput: Calculated by converting volumetric injection rates using assumed formation volume factors.
- Formation volume factors were assumed as follows:
 - CO₂ FVF = 0.5, based on typical values reported for reservoir conditions in Permian fields [14].
 - Water FVF = 1.0, assuming negligible compressibility for brine under reservoir pressure.

These parameters enabled the computation of an efficiency score for each scenario, defined as the ratio of retention to utilization. This score provides a quantitative basis for selecting optimal WAG ratios that balance CO₂ storage goals with oil recovery efficiency.

Methodology

This section describes the approach used to forecast short-term CO₂ and oil production responses in CO₂ EOR projects using the Temporal Fusion Transformer (TFT) model. The workflow consisted of five major stages: data preprocessing and normalization, parsing of CO₂ incremental response, model training, WAG scenario creation, and post-processing for performance and efficiency analysis.

Forecasting Model: Temporal Fusion Transformer (TFT)

The Temporal Fusion Transformer (TFT) is a state-of-the-art deep learning model for interpretable multi-horizon time series forecasting. Originally proposed by Lim et al. (2021), TFT combines long- and short-term memory components through LSTM layers with attention mechanisms, allowing it to capture both temporal dependencies and variable importance across time and feature dimensions.

TFT is particularly well-suited for CO₂ EOR forecasting due to its ability to incorporate static, time-varying known, and unknown covariates, its support for missing values and irregular time steps, and its scalability across multiple fields with differing temporal lengths.

Model implementation was done using the pytorch-forecasting library in Python with pytorch-lightning for training orchestration. All training was conducted on CPU to ensure reproducibility across environments.

Data Preprocessing

For each of the six fields, time series data was standardized using the following preprocessing steps:

- Time Re-indexing: All monthly data were aligned using a continuous time_idx variable starting from the onset of CO₂ injection (or miscible injection in SACROC).
- Scaling: The target variable (incremental EOR oil) was normalized using a StandardScaler, and scaled values were used during model training.
- Covariate Encoding: Time-varying known variables included:
 - BWIPD (barrels of water injected per day)
 - MCFID (thousand cubic feet of CO₂ injected per day)
 - Month number.

These covariates informed the model of known injection conditions at each timestep. Each field was treated independently, and group normalization was applied to the target variable.

Parsing CO₂ Response

The total oil production includes contributions from both waterflood and CO₂ injection. To isolate the incremental CO₂ response, an exponential decline model was fit to the pre-CO₂ waterflood period and extrapolated into the CO₂ injection period. The difference between the extrapolated curve and actual production was taken as the net CO₂-driven response. Figure 3 and Figure 4 shows an example of parsing out the incremental oil from DU and EV fields, respectively, using Exponential decline. It should be noted that EV field did not have as much data to fit the decline curve.

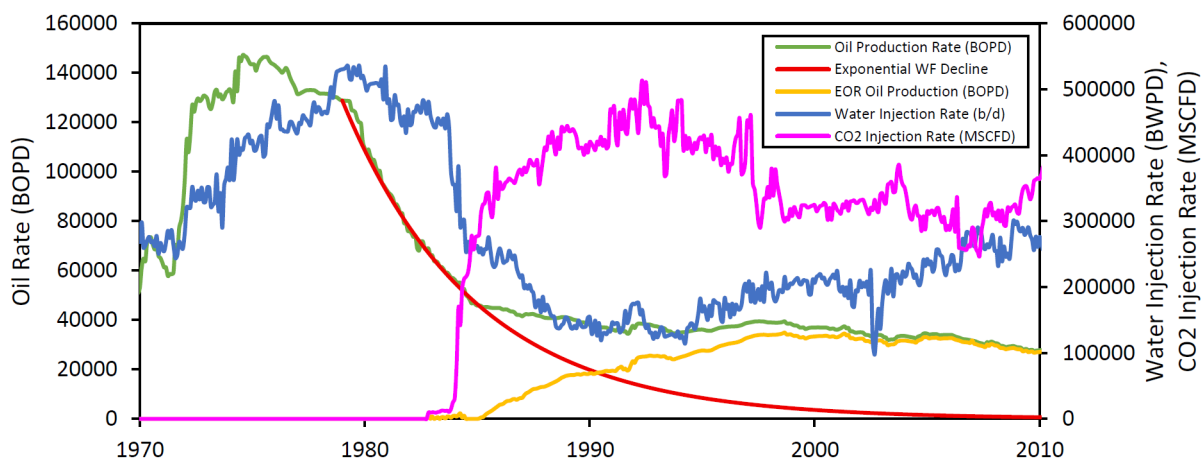


Figure 3—Example of EOR production extrapolation using decline curve analysis for DU field

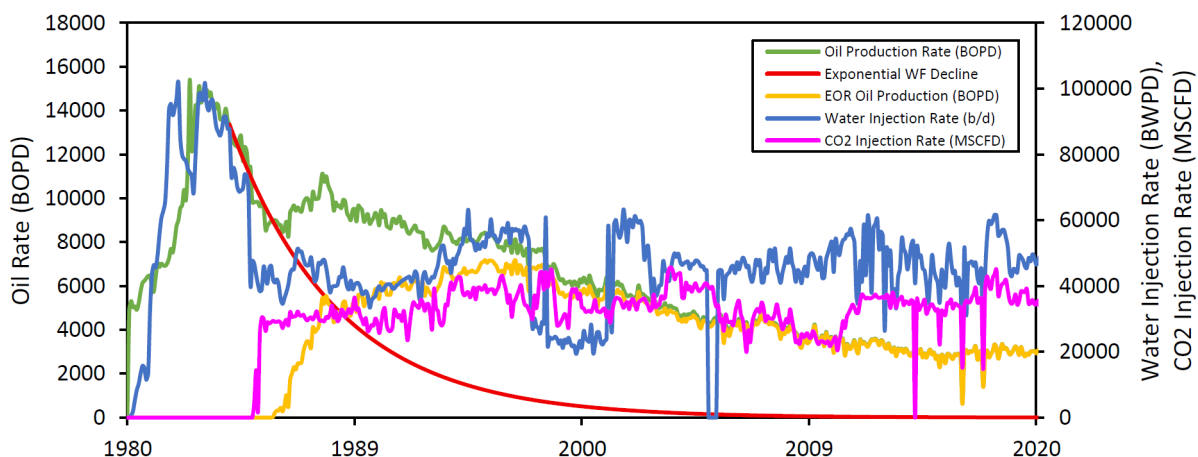


Figure 4—EOR production extrapolation using decline curve analysis for EV field

This approach is consistent with analytical best practices in field studies and ensures that the model focuses on forecasting CO₂-responsive oil rather than general field behavior. This parsed CO₂ EOR response became the model's forecast target (target_scaled).

Hyperparameter Optimization

To improve the performance of the Temporal Fusion Transformer (TFT) model, hyperparameter optimization was carried out to tune key parameters including max_encoder_length, hidden_size, attention_head_size, and learning_rate. A genetic algorithm-based search was used to explore the parameter space and minimize the validation root mean squared error (RMSE). This automated tuning process ensured better generalization and forecasting accuracy across all six CO₂ EOR fields. The selected hyperparameters were applied consistently across scenarios to preserve comparability of results.

Model Training and Validation

Each field's data was split into:

- Training set (80%)
- Validation set (10%)
- Test set (10%)

A "TimeSeriesDataSet" object was created per field using:

- max_encoder_length = 6 months
- max_prediction_length = 1 month (rolled iteratively for a 12-month horizon)
- Training used the following configuration:
 - Learning rate: 0.001
 - Hidden size: 64
 - Attention head size: 1
 - Dropout: 0
 - Loss function: RMSE
 - Early stopping: Patience of 200 epochs

Performance was evaluated using RMSE, MAE, and R^2 . Results showed strong training accuracy, particularly for EOR oil production, with average R^2 values exceeding 0.87 across fields. For example, [Figure 5](#) and [Figure 6](#) show the comparison between actual and model predicted incremental oil production and CO₂ production, respectively for DU field, while [Figure 7](#) and [Figure 8](#) show the oil and co₂ production for DU, respectively.

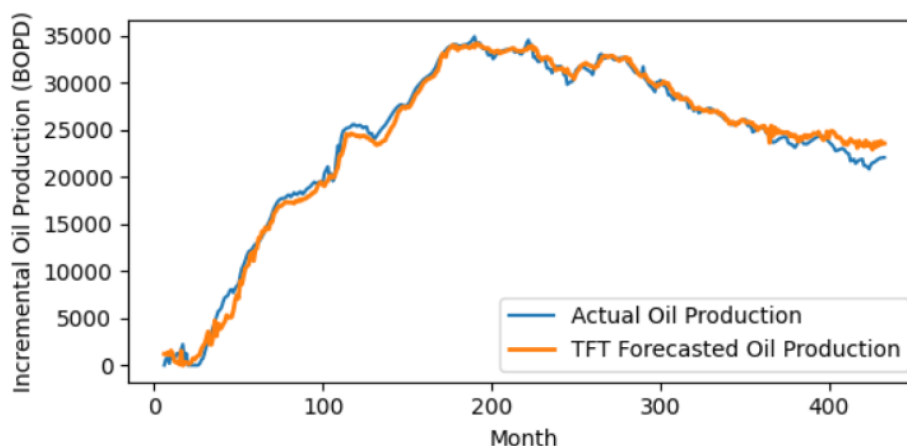


Figure 5—Actual vs Predicted Incremental Oil Production for DU

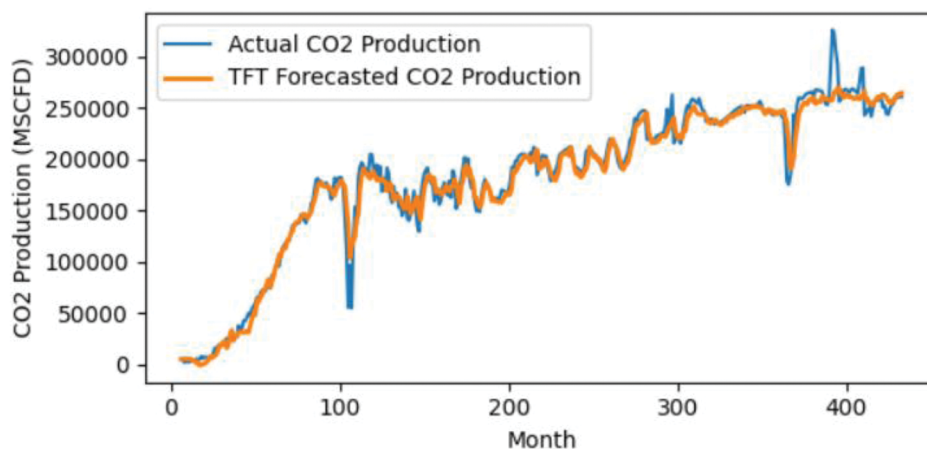


Figure 6—Actual vs Predicted CO2 Production for DU

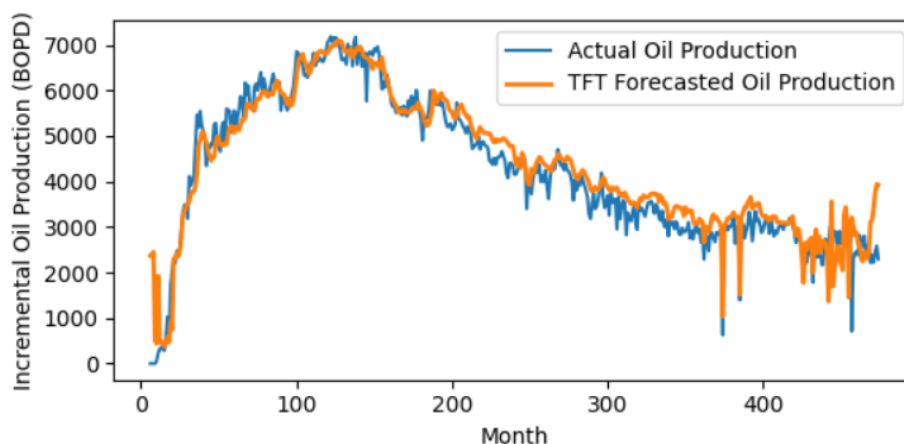


Figure 7—Actual vs Predicted Incremental Oil Production for EV

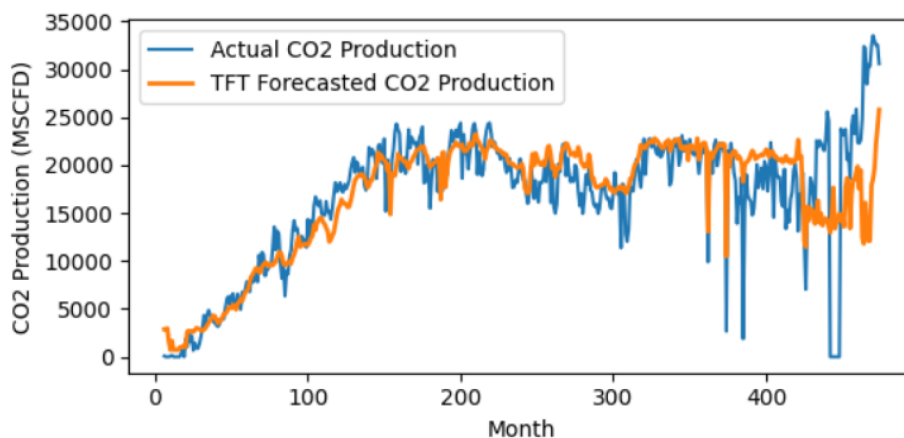


Figure 8—Actual vs Predicted CO2 Production for EV

WAG Scenario Design and Forecasting

To simulate the effect of different gas-water ratios (GWR) in WAG operations, five scenarios were developed per field using the GWR multipliers of $X = 1$ (base), 2, 3, 0.5, 0.33.

For each scenario, water and CO₂ injection rates were recomputed to maintain a constant injection volume and throughput while altering the gas/water ratio:

$$New\ CO_2\ Rate = 2 \frac{(0.5 \times C \times W)}{1 + \frac{1}{0.5 \times \frac{C}{W}}}$$

$$New\ Water\ Rate = 0.5 \times C + W - 0.5 \times New\ CO_2\ Rate$$

Where C and W are the original CO₂ and water injection rates, respectively. These modified rates were then input into the TFT model as future known covariates. Each scenario was forecasted for 12 months into the future, generating predictions for both CO₂ production (MCFPD) and EOR oil production (BOPD).

Separate models were trained for each target. For each forecast, the predicted oil and gas responses were used to compute:

- Utilization (MCF injected per BBL of oil)
- Retention (%)
- Efficiency score = Retention / Utilization

These metrics allowed quantitative ranking of WAG ratios.

Results and Discussion

Model Forecast Accuracy

The Temporal Fusion Transformer demonstrated high accuracy in forecasting both CO₂ and oil production across all six fields. Forecast quality was evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R²).

The model exhibited consistently strong performance in oil prediction, with R² values exceeding 0.87 in most fields and peaking at 0.95 in the Wasson San Andres (WSA) field. CO₂ production forecasts also showed excellent alignment, achieving R² values above 0.91 in five out of six fields, except for East Vacuum (EV), which recorded lower accuracy (R² = 0.54), likely due to greater operational noise and limited waterflood production data to fit the exponential decline to. Summary of accuracy metrics for both Oil and CO₂ production are shown in [Table 1](#) and [Table 2](#), respectively.

These results validate the applicability of TFT for short-term forecasting of CO₂ EOR responses, providing a robust alternative to full-physics simulation for annual planning.

Table 1—Forecast Accuracy Metrics for EOR Oil Production

Field	RMSE	MAE	R2	Relative Error (RMSE)
DU	3243	2586	0.87	12%
EV	505	389	0.90	16%
RWS	729	516	0.89	8%
SACROC	2205	1803	0.81	9%
SSAU	2635	2193	0.89	14%
WSA	2896	2085	0.95	7%

Table 2—Forecast Accuracy Metrics for CO₂ Production

Field	RMSE	MAE	R2	Relative Error (RMSE)
DU	21424	16011	0.91	9%
EV	4739	3266	0.54	21%
RWS	7081	5258	0.95	6%
SACROC	77816	67168	0.87	12%
SSAU	21523	16988	0.91	11%
WSA	37088	29727	0.92	9%

Optimization of WAG Ratios

To evaluate the operational impact of varying WAG ratios, five injection scenarios were simulated per field, adjusting the CO₂-to-water injection ratio (GWR) while holding total injection volume approximately constant. Key metrics defined earlier included utilization, retention and efficiency score. Figure 9 shows the efficiency trends for all fields with the colors corresponding to the GWR multiplier "X". It can be seen from the figure that there is room to optimize most of these fields since the base case (X=1) does not always exhibit the highest efficiency score. This data is also summarized in Table 3. A general pattern emerged: GWR = 2–3 produced the highest retention (up to 79%) but diminishing oil gains, leading to moderate efficiency scores. GWR = 0.3–0.5 offered better oil rates but negative retention due to continuation of production of pre-injected CO₂. An intermediate GWR of ~1.0–1.5 often balanced both goals, achieving retention of 50–60% with utilization rates between 13–21 MCF/BBL.

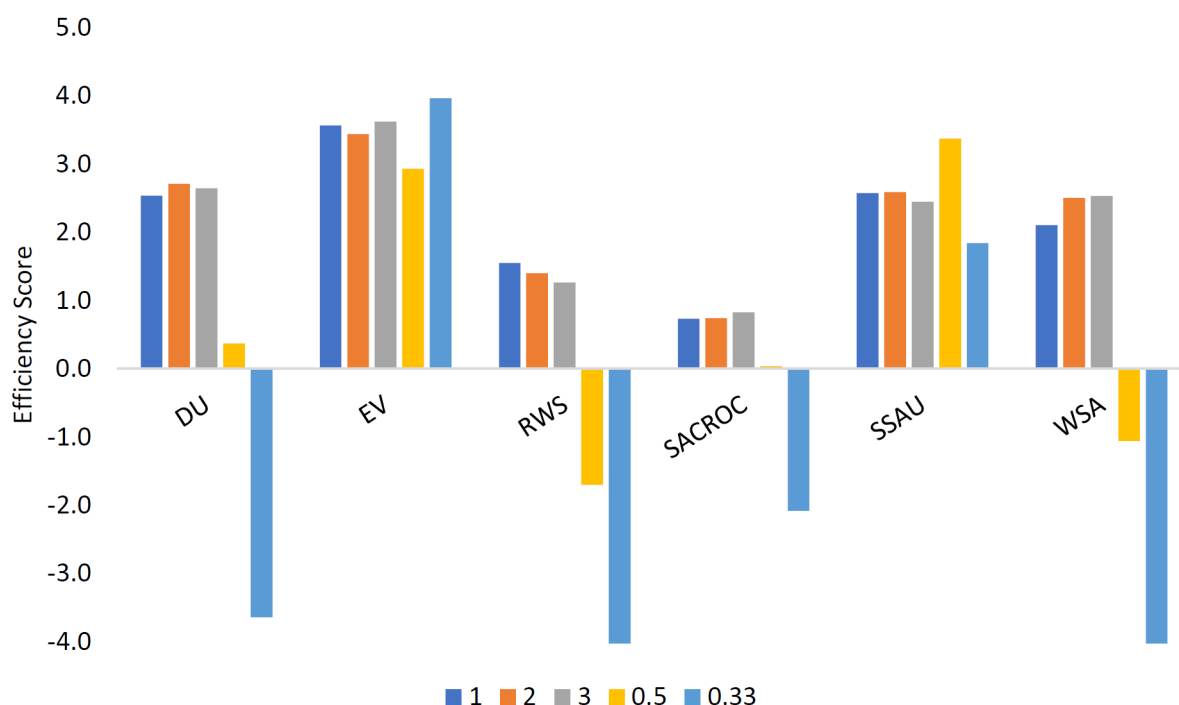


Figure 9—Efficiency scores for different fields at different WAG multiplier scenarios

Table 3—Summary of WAG Optimization Results

Field	GWR multiplier Scenario	GWR	Utilization (MCF/BBL)	CO2 Retention (%)	Efficiency Score
DU	1	0.9	15.2	39%	2.5
	2	1.7	21.1	57%	2.7
	3	2.6	23.7	63%	2.6
	0.5	0.4	10.6	4%	0.4
	0.33	0.3	8.5	-31%	-3.6
EV	1	0.7	14.3	51%	3.6
	2	1.4	19.6	67%	3.4
	3	2.0	21.7	79%	3.6
	0.5	0.3	9.5	28%	2.9
	0.33	0.2	6.8	27%	4.0
RWS	1	0.4	19.3	30%	1.5
	2	0.7	36.0	50%	1.4
	3	1.1	47.7	60%	1.3
	0.5	0.2	11.6	-20%	-1.7
	0.33	0.1	8.0	-68%	-8.5
SACROC	1	0.6	33.8	25%	0.7
	2	1.3	50.1	37%	0.7
	3	1.9	58.8	49%	0.8
	0.5	0.3	21.8	1%	0.0
	0.33	0.2	19.0	-39%	-2.1
SSAU	1	0.9	20.5	53%	2.6
	2	1.9	24.7	64%	2.6
	3	2.8	28.0	68%	2.4
	0.5	0.5	11.9	40%	3.4
	0.33	0.3	9.4	17%	1.8
WSA	1	1.1	13.8	29%	2.1
	2	2.1	18.0	45%	2.5
	3	3.2	20.0	51%	2.5
	0.5	0.5	9.1	-10%	-1.1
	0.33	0.3	6.8	-51%	-7.6

Table 4 presents the recommended adjustments to the gas-water ratio (GWR) for each field based on short-term Temporal Fusion Transformer (TFT) forecasts. The goal was to maximize CO₂ sequestration efficiency, defined here as the ratio of CO₂ retention to utilization (MCF injected per barrel of oil produced). For most fields, adjusting the GWR yields measurable improvements in efficiency. In the Denver Unit (DU), increasing the GWR from 0.9 to 1.7 results in an 8% gain in efficiency, with CO₂ utilization improving by 38% and retention by 48%. SACROC shows a modest efficiency gain (+14%) when increasing the GWR to 1.9, driven by significant improvements in retention (+96%). Conversely, fields such as SSAU and EV benefit from reducing the GWR. In SSAU, decreasing the GWR from 0.9 to 0.5 leads to the largest relative improvement in efficiency (+31%), despite reductions in both utilization and retention. For

EV, reducing the GWR to 0.2 increases efficiency by 11%, though model uncertainty (low R^2) warrants cautious interpretation. Fields like RWS show minimal sensitivity to GWR changes within the tested range, suggesting that other operational levers may be more impactful. Meanwhile, WSA shows that pushing to higher GWR values (3.2) can yield a 19% efficiency improvement, albeit with a notable increase in CO₂ injection rate.

Table 4—Optimization recommendations and simulated benefits of the study

Field	Current		Recommended			Benefit		
	GWR	Efficiency Score	GWR	Efficiency Score	Δ CO ₂ Injection Rate (MMCFPD)	Δ Efficiency (%)	Δ Utilization (%)	Δ CO ₂ Retention (%)
DU	0.9	2.5	1.7	2.7	150	+8%	+38%	+48%
EV*	0.7	3.6	0.2	4	-25	+11%	-52%	-47%
RWS	0.4	1.5	0.4	1.5	0	0%	0%	0%
SACROC	0.6	0.7	1.9	0.8	604	+14%	+74%	+96%
SSAU	0.9	2.6	0.5	3.4	-135	+31%	-42%	-24%
WSA	1.1	2.1	3.2	2.5	265	+19%	+45%	+75%

Note:

* EV results should be interpreted with caution due to lower model accuracy ($R^2 < 0.6$) during training.

These findings support field-specific, short-term WAG adjustments as a practical lever for improving CO₂ management performance in mature EOR assets.

Operational Insights

From the modeling exercise, several operational insights emerged. Short-term optimizations (12-month horizon) can be meaningfully supported with machine learning tools that are data-driven and less dependent on static reservoir models. Additionally, the efficiency score proposed here enables simultaneous targeting of both climate (sequestration) and economic (oil recovery) KPIs. This study shows that deep learning forecasting tools, particularly TFT, can play a critical role in bridging production optimization and carbon management. The efficiency score developed here—linking CO₂ retention and utilization—serves as a proxy for carbon abatement effectiveness. By embedding such models into operational workflows, operators can make evidence-based adjustments to WAG ratios, directly enhancing sequestration permanence while maintaining oil revenue streams. These insights are especially useful for compliance-driven projects targeting carbon credit monetization under programs such as 45Q and LCFS.

Conclusions and Recommendations

This study demonstrates the successful application of a Temporal Fusion Transformer (TFT) deep learning model to forecast CO₂ and oil production performance across six legacy CO₂ EOR fields under a range of Water-Alternating-Gas (WAG) ratios. By digitizing historical injection and production data and applying consistent forecasting and optimization workflow, we achieved several key outcomes:

Key Conclusions

The TFT model achieved high predictive accuracy across diverse field conditions, with R^2 values exceeding 0.87 for oil production and 0.91 for CO₂ production in most cases. This confirms the model's suitability for short-term operational forecasting without detailed reservoir models.

EV field showed a low R^2 fit likely due to poor quality of EOR oil extrapolation using exponential decline. This is due to the lack of extended period of waterflood decline oil production. This shows that a good exponential fit and extrapolation is essential in properly training the TFT model and accurately predicting future performance

A clear tradeoff emerged between CO_2 utilization (MCF per barrel) and CO_2 retention (%) as WAG ratio increased. While higher GWRs improved retention, they often led to diminishing returns in oil recovery. Conversely, low GWRs reduced CO_2 storage effectiveness and, in some cases, resulted in negative retention.

Optimal WAG ratios varied by field, with the best-performing scenarios typically falling in the GWR range of 1.5–2.5, balancing both oil recovery and sequestration goals. The efficiency score proposed in this study (Retention / Utilization) provides a useful single-parameter screening tool to guide such optimization.

To the authors' knowledge, this is the first published study applying the TFT architecture to multivariate CO_2 EOR forecasting across multiple fields and using it to evaluate short-term WAG strategy optimization. Unlike traditional simulation approaches, this workflow enables rapid scenario testing using only field-level production data.

Adopting TFT-based forecasts can help reduce CO_2 utilization by up to 42% and increase retention by as much as 96% (e.g., in SACROC), providing tangible gains in sequestration efficiency.

Recommendations

The proposed framework can be integrated into annual business planning and reservoir review cycles, providing low-latency forecasts and rapid screening of WAG scenarios. This supports timely decision-making in mature CO_2 EOR operations.

While model accuracy was strong overall, future work should incorporate additional variables (e.g., injection pressures, well spacing, tracer data) to improve accuracy in more heterogeneous settings like East Vacuum.

A hybrid approach combining TFT-based forecasting with physics-based simulation could offer the best of both worlds: speed, interpretability, and physical rigor—especially for life-of-field strategy design and regulatory assessments.

The use of machine learning to improve short-term CO_2 retention and reduce utilization rates can support net-zero and low-carbon targets, especially when coupled with carbon credit frameworks. This highlights the role of data science not just in EOR, but in CCUS strategy optimization more broadly.

Disclaimer

The analyses and recommendations presented in this paper are intended solely for research and educational purposes. They are based on publicly available data, machine learning forecasts, and simplified operational assumptions. They are not intended to substitute for detailed reservoir studies or engineering judgment. Field implementation should be preceded by appropriate technical due diligence, risk assessment, and consultation with qualified professionals.

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